Understanding the Relationship Between Rodent Violations and Extreme Temperatures

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Research Question:

What is the relationship between the number of rodent-related health code violations and the frequency of heat advisory warnings (extreme summer temperatures) in Boston from 2007-2025?

Background:

- Increase of extreme temperatures during summer due to climate change.
- Rats actively avoid extreme heat, seeking shelter in human spaces during this time.
- This causes restaurants to be at risk, as they have an abundance of food, water, and shelter.

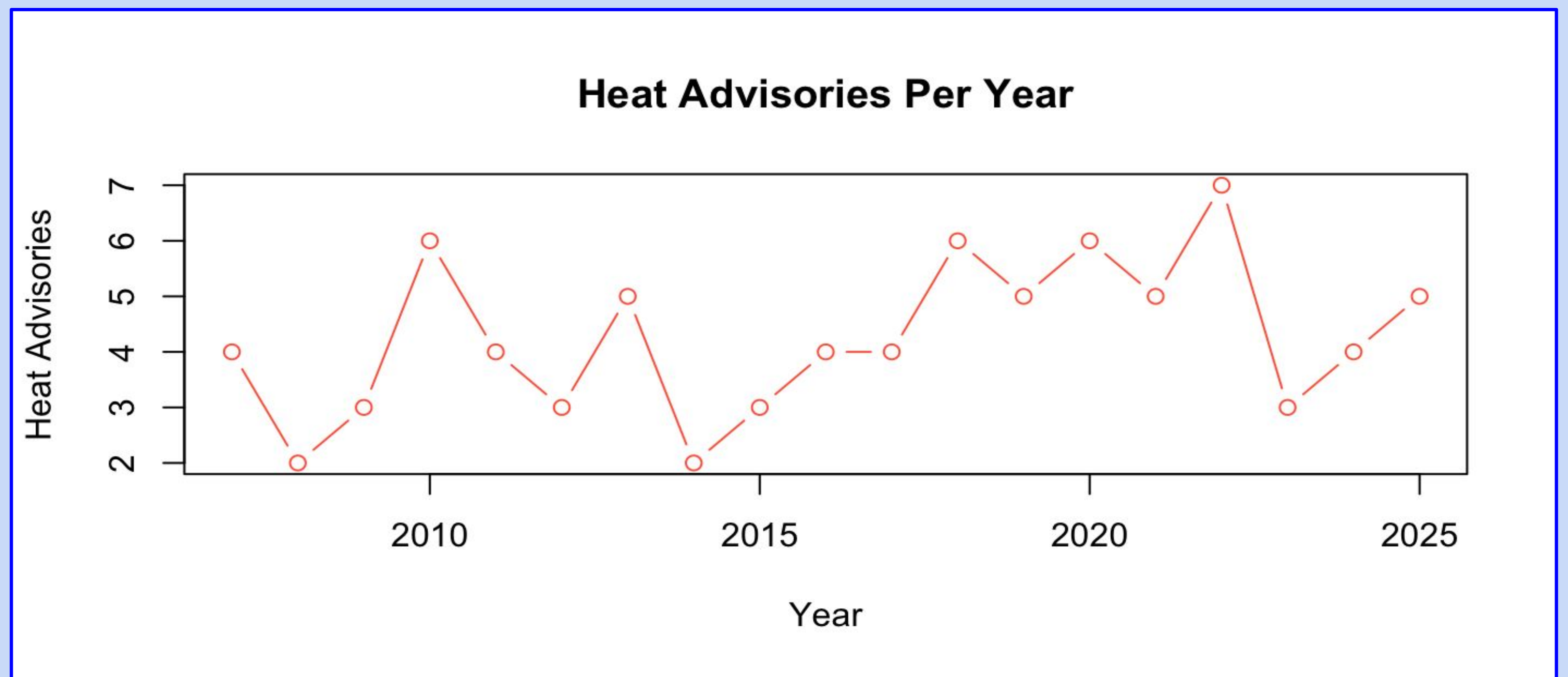

Figure 1: Scatterplot of heat waves, categorized by 3 consecutive days of 85 degrees of more.

Data & Methods:

Primary Data Source:

- Boston Food Inspections
  - Attributes: violation time, comments
- NOAA Climate Data for Boston
  - Attributes: max temp

Data Manipulation:

- Created rodent_rate (total violations in a day / total inspections in a day)
  - Characterized by “rodent” in comments
- Created heat_wave_start (if it was the beginning of a heat advisory – 3 days of 85° or more)

Visualizations:

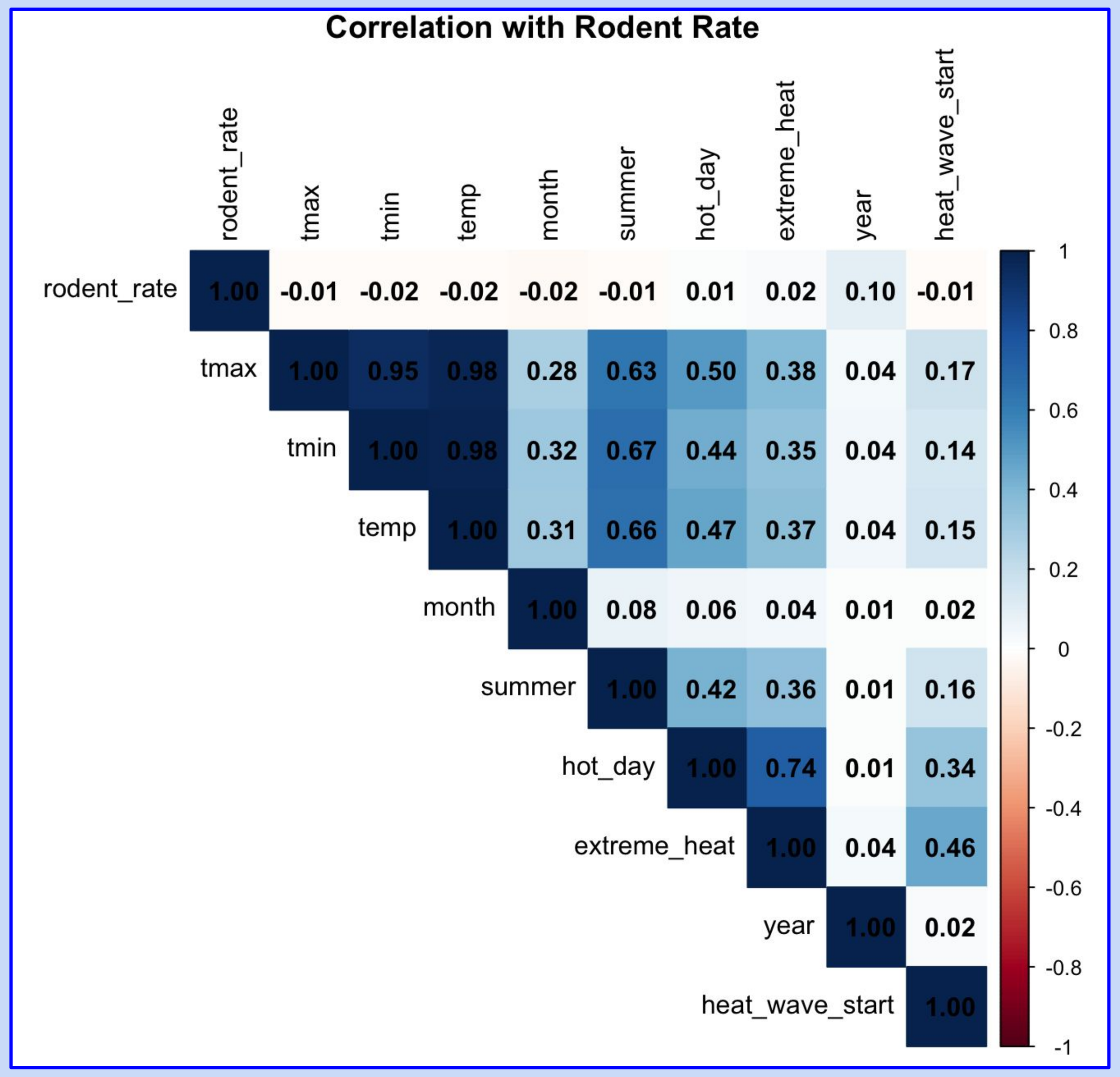

Figure 2: Correlation plot of all potential variables

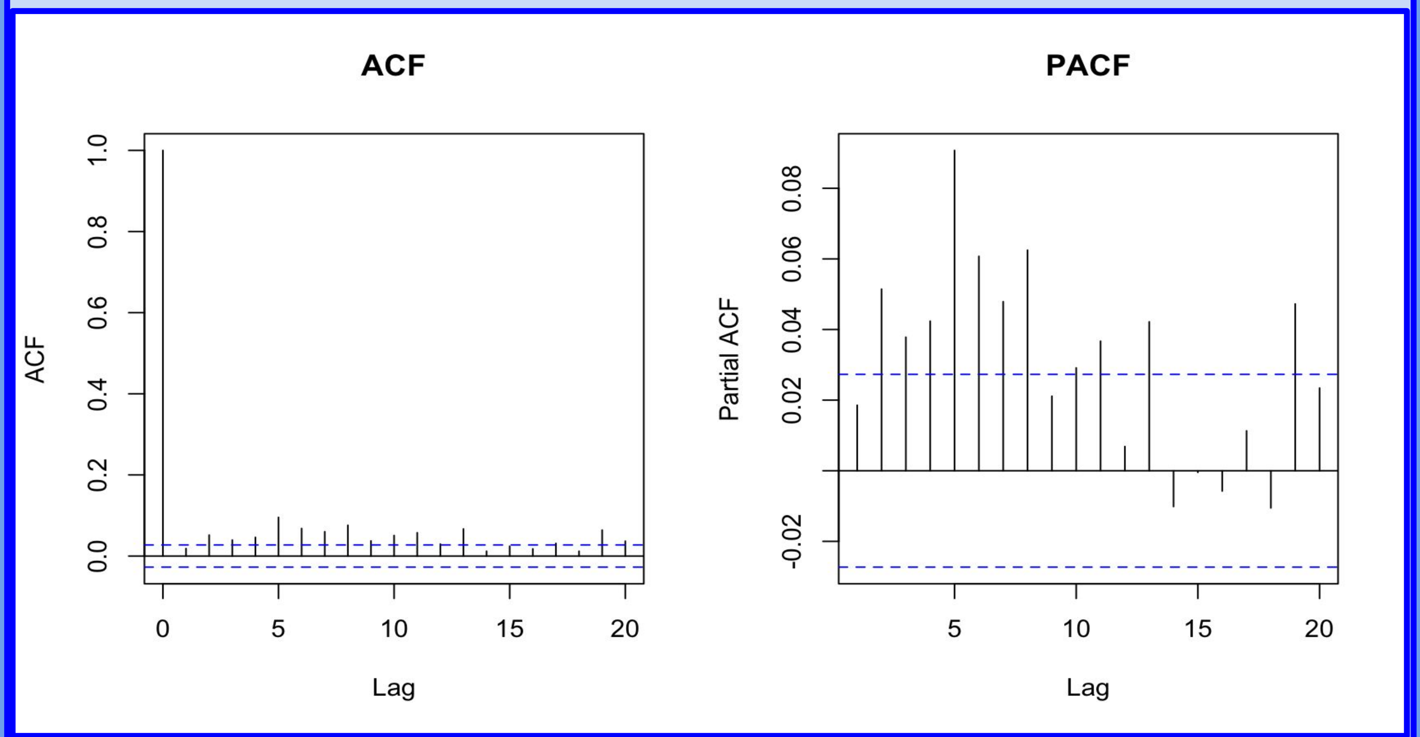

Figure 3: ACF and PACF plots

Modeling:

- Plotted ACF & PACF plots to determine potential lags and stationarity in data
  - Appeared to be no clear autocorrelation

Model 1: OLS Regression with no autocorrelation

$$\text{Rodent Rate}_t = \beta_0 + \beta_1 \text{Date}_t + \beta_2 \text{Tmax}_t + \beta_3 \text{Hot Day}_t + \beta_4 \text{Extreme Heat}_t + \beta_5 \text{Heat Wave Start}_t + \epsilon_t$$

Model 2:

Part 1: Beta Regression – Given that violations actually occurred, what rodent rate can be predicted?

$$\text{logit}\left(P(\text{Any Rodent Violation}_t)\right) = \beta_0 + \beta_1 \text{Date}_t + \beta_2 \text{Tmax}_t + \beta_3 \text{Heat Wave Start}_t$$

Part 2: Logistic Regression – Any rodent violations on a specific day?

$$\text{logit}(\mu_t) = \beta_0 + \beta_1 \text{Date}_t + \beta_2 \text{Tmax}_t + \beta_3 \text{Heat Wave Start}_t \qquad \mu_t = E(\text{Rodent Rate}_t)$$

Results:

Model 2 Pt. 1:

- **Date** – 1.000146 (p-value < 0.001)

For each additional day in the study period, odds of a violation increase by **0.015%**

- **Tmax** – 0.99 (p-value = 0.003)

For each 1°F increase in max temperature, odds of any violation decrease by **~1%**

- **Heat_wave_start** – 1.012 (not significant)

Essentially no effect — odds increase by just 1.2% at heat wave start

Model 2 Pt. 2:

No time trend, max temp, nor heat wave influence the rate of rodent violations

Conclusion: Heat waves **do not appear** to drive rodent violation rates.

- Notable: The **negative tmax effect in Part 1**; This suggests that inspections may be suppressed on hot days